How-To Library

Informatica

Data Engineering Integration 10.4.0 on AWS Databricks: Performance Tests

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Abstract

Customers of Amazon Web Services and Informatica® can deploy Informatica Data Engineering Integration in the AWS cloud platform to run mappings on the Databricks compute cluster. Auto-scaling is an appropriate approach for many cases, but also comes at a cost during the initial mapping run. This article describes results of performance tests on a Databricks cluster, using a range of data volumes and cluster sizes.

Supported Versions

- Data Engineering Integration 10.4.0

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Overview

The following sections contain findings from tests on a Data Engineering Integration deployment using a Databricks cluster on AWS.

Sources and targets were stored on S3 storage. Except where noted, the tests were conducted on an m4.2xlarge Databricks cluster with the following configuration:

- 32 GB RAM
- 8 cores
- 1.5 DBU

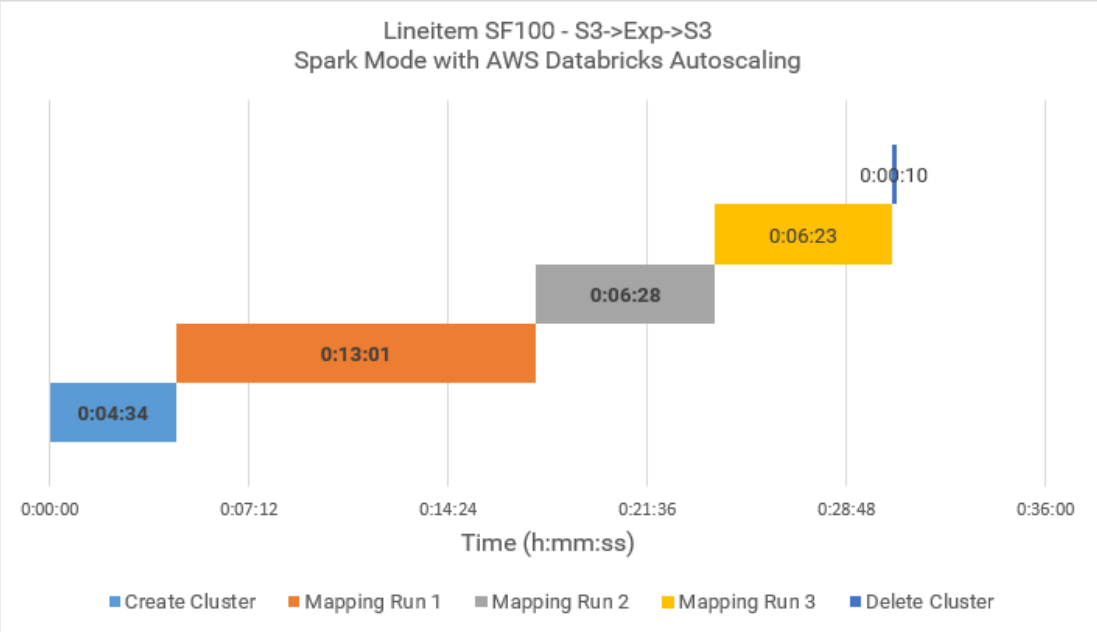
  Note: A Databricks Unit (DBU) is a unit of processing capability per hour. One DBU is the processing capability of one hour of one r3.xlarge (memory optimized) or one c3.2xlarge (compute optimized) AWS instance. For more information, see the Databricks documentation.

Performance of a Mapping with Auto-Scaling

This test measured the time it takes to run a mapping on an auto-scaled Databricks cluster on AWS.

The mapping processed a data volume of 100 GB. The test was run on an environment of between 2 and 16 nodes running on m4.2xlarge AWS EC2 instances, each with 8 cores and 32MB memory.

The following image shows elapsed time for the mapping run:

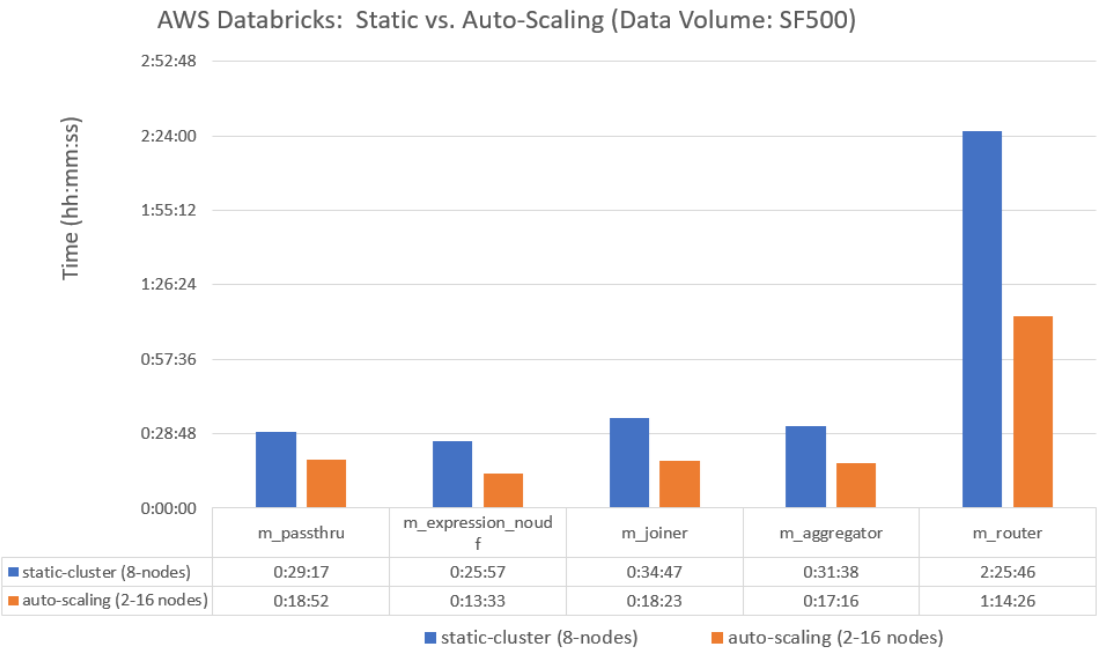


Notice that, after cluster creation, the first mapping run takes significantly longer than subsequent runs. This relatively longer time elapses as the cluster spins up the additional nodes required to run the job. The run time is less for subsequent runs when all the nodes are available.

Performance: Static and Auto-Scaled Clusters

Mappings in this test processed a data volume of 500 GB. The test compared the performance of auto-scaled clusters with between 2 and 16 nodes against static clusters with 8 nodes.

The following image shows test results:



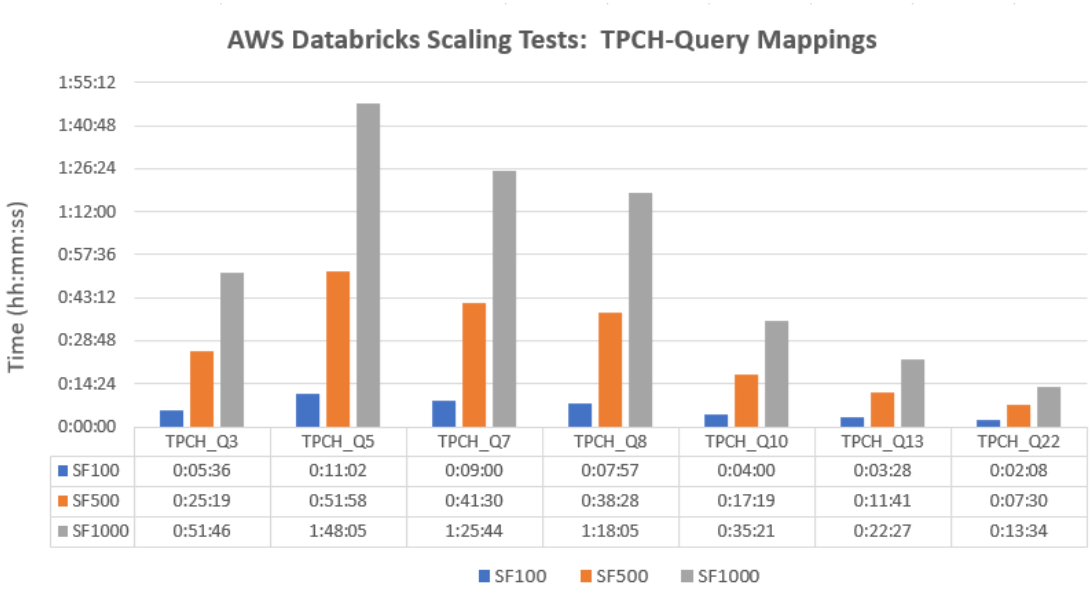
The test found that mappings ran between 36-49% faster on auto-scaled clusters when a large number of nodes were spun up. However, note the following guidelines:

- Smaller jobs require fewer additional nodes, so the performance gains of auto-scaling are less significant.
- A static cluster tends to be more costly than an auto-scaled cluster because all nodes are constantly running even when they are not required for job processing.

Data Volume Scaling Test

This test ran TPCH queries comparing performance using three different data volumes.

The following image shows results of the test:



	TPCH_Q3	TPCH_Q5	TPCH_Q7	TPCH_Q8	TPCH_Q10	TPCH_Q13	TPCH_Q22
SF100	0:05:36	0:11:02	0:09:00	0:07:57	0:04:00	0:03:28	0:02:08
SF500	0:25:19	0:51:58	0:41:30	0:38:28	0:17:19	0:11:41	0:07:30
SF1000	0:51:46	1:48:05	1:25:44	1:18:05	0:35:21	0:22:27	0:13:34

The test showed that the time it takes to run jobs increases in a linear pattern, depending on data volume, with a 1:1 correspondence between data volume and elapsed time.

Author

Mark Pritchard